

Chapter 2: Limits

Limits of Sequences & Limits of Functions

Advanced Mathematics
(Tongji University, 8th Ed.)

Outline

1 Limits of Sequences

2 Limits of Functions

Section 2: Limits of Sequences — Roadmap

- ① Definition of the Limit of a Sequence
- ② Properties of Convergent Sequences
- ③ Existence Criteria and Two Important Limits

Key Idea

A sequence $\{x_n\}$ **converges** to a if its terms get arbitrarily close to a as n grows.

Motivation: Measuring Area

Problem

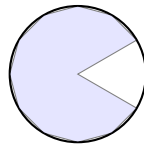
Find the area of a region by inscribing polygons.

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- True value of side length: unknown.



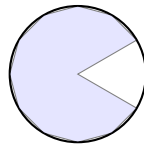
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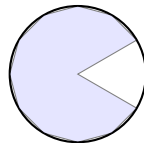
Inscribed polygon

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Problem

Find the area of a region by inscribing polygons.

- True value of side length: unknown.
- Direct observation gives approximate values.
- As precision improves, approximations approach the true value.



Inscribed polygon

What Is a Sequence?

Definition (Sequence)

If the independent variable takes positive integer values, the function

$$x_n = f(n), \quad n = 1, 2, 3, \dots$$

is called a **sequence**, denoted by $\{x_n\}$.

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Terminology

- x_n is called the **general term** (or n -th term).
- The sequence is also written as $x_1, x_2, x_3, \dots, x_n, \dots$

ε - N Definition of Limit

Definition (ε - N Definition)

Let $\{x_n\}$ be a sequence and a a constant. If for every $\varepsilon > 0$, there exists a positive integer N such that

$$n > N \implies |x_n - a| < \varepsilon,$$

then we say the sequence **converges** to a , denoted by

$$\lim_{n \rightarrow \infty} x_n = a \quad \text{or} \quad x_n \rightarrow a \quad (n \rightarrow \infty).$$

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Divergence

If no such a exists, the sequence is said to **diverge**.

Geometric Interpretation of ε - N

Definition: independent variable take positive integer's function is called sequence $x_n = f(n)$ or $\{x_n\}$. x_n is called 通项 (general term) denoted by x_n (term).

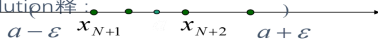
if sequence $\{x_n\}$ 及常数 a 有下列关系:

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then称 $\{x_n\}$ 的极限为 a , denoted by $\lim_{n \rightarrow \infty} x_n = a$ or $x_n \rightarrow a (n \rightarrow \infty)$

this when also 称 sequence converge, otherwise 称 sequence diverge.

geometric interpretation:



$$a - \varepsilon < x_n < a + \varepsilon$$

即当 $n > N$ 时, 有

$$x_n \in \cup(a, \varepsilon)$$

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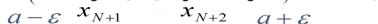
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Equivalent Formulation

$$\lim_{n \rightarrow \infty} x_n = a \iff \forall \varepsilon > 0, \exists N > 0: n > N \Rightarrow x_n \in (a - \varepsilon, a + \varepsilon).$$

Convergent vs. Divergent Sequences

Convergent

- $\frac{1}{n} \rightarrow 0$
- $\frac{n}{n+1} \rightarrow 1$
- $q^n \rightarrow 0$ ($|q| < 1$)
- $\frac{(-1)^n}{n} \rightarrow 0$

Divergent

- $(-1)^n$: oscillates
- $n^2 \rightarrow +\infty$
- $(-n) \rightarrow -\infty$
- $1 + (-1)^n/n$: bounded but not convergent

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Trend is definite $\Rightarrow$ convergent; trend is indefinite $\Rightarrow$ divergent.

Example 1: $\lim_{n \rightarrow \infty} \frac{n + (-1)^n}{n} = 1$

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Proof Sketch.

Compute:

$$|x_n - 1| = \left| \frac{n + (-1)^n}{n} - 1 \right| = \left| \frac{(-1)^n}{n} \right| = \frac{1}{n}.$$



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Finding N

For any $\varepsilon > 0$, we want $|x_n - 1| < \varepsilon$, i.e. $\frac{1}{n} < \varepsilon$; this holds as long as $n > \frac{1}{\varepsilon}$.

Example 1 (continued)

Conclusion

Take $N = \left\lfloor \frac{1}{\varepsilon} \right\rfloor$. Then whenever $n > N$,

$$\left| \frac{n + (-1)^n}{n} - 1 \right| = \frac{1}{n} < \varepsilon.$$

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Take $N = \left\lfloor \frac{1}{\varepsilon} \right\rfloor$. Then whenever $n > N$,

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Example 2: $\lim_{n \rightarrow \infty} \frac{(-1)^n}{(n+1)^2} = 0$

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$$|x_n - 0| = \left| \frac{(-1)^n}{(n+1)^2} \right| = \frac{1}{(n+1)^2} < \frac{1}{n+1}.$$



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Finding N

For any $\varepsilon \in (0, 1)$, want $|x_n - 0| < \varepsilon$.

It suffices that $\frac{1}{n+1} < \varepsilon$, i.e. $n > \frac{1}{\varepsilon} - 1$.

Example 2 (continued)

Conclusion

Take $N = \left\lfloor \frac{1}{\varepsilon} - 1 \right\rfloor$. Then when $n > N$,

$$|x_n - 0| < \varepsilon \implies \lim_{n \rightarrow \infty} \frac{(-1)^n}{(n+1)^2} = 0.$$

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Note

N depends on ε but is **not unique**.

One could also take $N = \left\lfloor \frac{1}{\sqrt{\varepsilon}} - 1 \right\rfloor$ (a tighter bound).

Any N satisfying the condition works; it need not be the smallest.

Example 3: $\lim_{n \rightarrow \infty} \frac{1}{n^\alpha} = 0 \ (\alpha > 0)$

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This is equivalent to $n^\alpha > \frac{1}{\varepsilon}$, i.e. $n > \varepsilon^{-1/\alpha}$. □

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Conclusion

Take $N = \lfloor \varepsilon^{-1/\alpha} \rfloor$. Then $n > N \implies \frac{1}{n^\alpha} < \varepsilon$. Hence $\lim_{n \rightarrow \infty} \frac{1}{n^\alpha} = 0$. ■

Property 1: Uniqueness of the Limit

Theorem (Uniqueness)

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Property 1 (continued)

Proof (cont.)

Let $N = \max\{N_1, N_2\}$. When $n > N$:

$$\begin{aligned}|a - b| &= |(a - x_n) + (x_n - b)| \\ &\leq |x_n - a| + |x_n - b| \\ &< \varepsilon + \varepsilon = 2\varepsilon = |a - b|.\end{aligned}$$

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Therefore the limit must be unique. ■

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But x_n alternates between 1 and -1 :

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Contradiction $\Rightarrow \{(-1)^n\}$ **diverges**. ■

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Let $\lim_{n \rightarrow \infty} x_n = a$. Take $\varepsilon = 1$. Then $\exists N$: for all $n > N$, $|x_n - a| < 1$. Hence $|x_n| < |a| + 1$ for $n > N$.

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Let $M = \max\{|x_1|, |x_2|, \dots, |x_N|, |a| + 1\}$. Then $|x_n| \leq M$ for all n . □

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Therefore $\{x_n\}$ is bounded. ■

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Boundedness does **not** imply convergence.

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Example (Counterexample)

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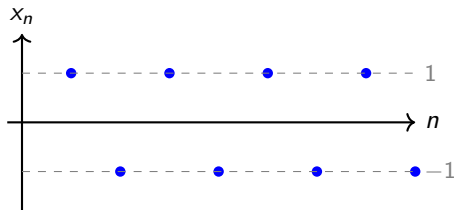
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Bounded but oscillating — no limit

Property 3: Local Sign-Preserving Property

Theorem (Sign-Preserving)

If $\lim_{n \rightarrow \infty} x_n = a$ and $a > 0$ (or $a < 0$), then from some term onward, $x_n > 0$ (or $x_n < 0$).

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In particular, $x_n > a - \frac{a}{2} = \frac{a}{2} > 0$ for all $n > N$. □

Corollary: Strictly Positive Limit

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Stronger Form

More generally, if $a > b$, then $x_n > b$ for all sufficiently large n . (Apply the theorem to the sequence $y_n = x_n - b$.)

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The subsequence indices satisfy $n_k \geq k$. Choose $K = N$. Then for all $k > K$:

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$$n_k \geq k > N \implies |x_{n_k} - a| < \varepsilon.$$

Hence $\lim_{k \rightarrow \infty} x_{n_k} = a$. ■

Using Subsequences to Prove Divergence

Practical Criterion for Divergence

If a sequence has two subsequences converging to **different** limits, then the original sequence **diverges**.

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Example

$$x_n = \sin\left(\frac{n\pi}{2}\right):$$

- Subsequence $x_{4k} = \sin(2k\pi) = 0 \rightarrow 0$
- Subsequence $x_{4k+1} = \sin(2k\pi + \frac{\pi}{2}) = 1 \rightarrow 1$

Different limits $\Rightarrow \{x_n\}$ diverges.

Existence Criteria: Overview

How do we know a limit exists without knowing its value?

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❶ Squeeze (Sandwich) Criterion

(p. 46)

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Key Insight

These criteria allow us to prove existence of limits even when the exact value is hard to compute directly.

Criterion 1: Squeeze (Sandwich) Theorem

Theorem (Squeeze Criterion)

Suppose sequences $\{y_n\}$, $\{x_n\}$, $\{z_n\}$ satisfy:

(1) $y_n \leq x_n \leq z_n$ for $n = 1, 2, \dots$

(2) $\lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} z_n = a$

Then $\lim_{n \rightarrow \infty} x_n = a$.

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Then $\lim_{n \rightarrow \infty} x_n = a$.

1. squeeze criterion

(criterion1) (p46)

$$\left. \begin{array}{l} (1) \quad y_n \leq x_n \leq z_n \quad (n = 1, 2, \dots) \\ (2) \quad \lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} z_n = a \end{array} \right\} \implies \lim_{n \rightarrow \infty} x_n = a$$

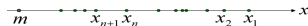
2. monotone bounded sequence must

has limit (criterion2) (p48)

$$m \leq x_1 \leq x_2 \leq \dots \leq x_n \leq x_{n+1} \leq \dots \leq M \implies \lim_{n \rightarrow \infty} x_n = a \quad (\leq M)$$



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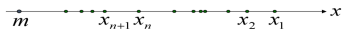
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$$\text{Monotone} + \text{Bounded} \implies \lim_{n \rightarrow \infty} x_n = b \quad (\leq M \text{ or } \geq m)$$

*Criterion 3: Cauchy Convergence Principle

Theorem (Cauchy Criterion)

A sequence $\{x_n\}$ has a limit **if and only if**:

$$\forall \varepsilon > 0, \exists N > 0 : m, n > N \implies |x_m - x_n| < \varepsilon.$$

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“Sufficiency” ($\Leftarrow$)

Proof omitted (requires completeness of $\mathbb{R}$).

Summary: Limits of Sequences

- ① ε - N **Definition** and applications
- ② **Properties of convergent sequences:**
 - Uniqueness
 - Boundedness (converse false!)
 - Local sign-preserving property
 - All subsequences converge to the same limit
- ③ **Existence Criteria:**
 - Squeeze (Sandwich) Criterion
 - Monotone Bounded Criterion
 - *Cauchy Convergence Principle

Discussion Question

What are the ways to show that a limit does **not** exist?

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What are the ways to show that a limit does **not** exist?

- 1 Find a subsequence tending to $+\infty$ or $-\infty$.
- 2 Find two subsequences converging to **different** limits.

Section 3: Limits of Functions — Roadmap

- ① Limits as $x \rightarrow x_0$ (finite point)
- ② Limits as $x \rightarrow \infty$ (at infinity)

Central Question

What does it mean for $f(x)$ to approach a value A as x approaches some target?

Motivation: Precision in Measurement

Problem

Measure the area of a square by observing its side length.

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Key Question

Given any precision $\varepsilon > 0$, how close must x be to x_0 to guarantee $|x^2 - x_0^2| < \varepsilon$?

Answer: find $\delta > 0$ such that $|x - x_0| < \delta$ suffices.

Definition 1: ε - δ Definition ($x \rightarrow x_0$)

Definition (ε - δ Limit)

Let $f(x)$ be defined in some deleted neighbourhood of x_0 . If

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x - x_0| < \delta \implies |f(x) - A| < \varepsilon,$$

then the constant A is called the **limit** of $f(x)$ as $x \rightarrow x_0$:

$$\lim_{x \rightarrow x_0} f(x) = A \quad \text{or} \quad f(x) \rightarrow A \quad (x \rightarrow x_0).$$

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Equivalent Formulation

$$\lim_{x \rightarrow x_0} f(x) = A \iff \forall \varepsilon > 0, \exists \delta > 0 : x \in \dot{U}(x_0, \delta) \implies |f(x) - A| < \varepsilon.$$

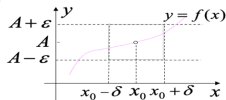
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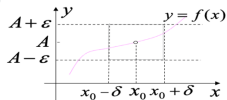
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geometric interpretation



this shows:

limit existence $\implies$ function locally bounded
 (P32 Theorem 2)

Immediate Consequence

Limit existence $\implies$ function is **locally bounded** near x_0 .

(P32, Theorem 2)

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Prove $\lim_{x \rightarrow x_0} C = C$ where C is a constant.

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Prove $\lim_{x \rightarrow x_0} x^2 = x_0^2$ using the ε - δ definition.

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Choosing δ

For given $\varepsilon > 0$, take $\delta = \min \left\{ 1, \frac{\varepsilon}{2|x_0| + 1} \right\}$.

Then $0 < |x - x_0| < \delta \implies |x^2 - x_0^2| < \varepsilon$. ■

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Proof.

$$|f(x) - A| = |(2x - 1) - 1| = 2|x - 1|.$$

For any $\varepsilon > 0$, we want $|f(x) - A| < \varepsilon$.

This holds as long as $|x - 1| < \frac{\varepsilon}{2}$. □

Example 2 (continued)

Conclusion

Take $\delta = \frac{\varepsilon}{2}$. Then when $0 < |x - 1| < \delta$:

$$|f(x) - A| = |(2x - 1) - 1| < \varepsilon.$$

Therefore $\lim_{x \rightarrow 1} (2x - 1) = 1$. ■

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For any $\varepsilon > 0$, take $\delta = \varepsilon\sqrt{x_0}$.

Then $0 < |x - x_0| < \delta$ implies $|\sqrt{x} - \sqrt{x_0}| < \varepsilon$.

Hence $\lim_{x \rightarrow x_0} \sqrt{x} = \sqrt{x_0}$. 

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Prove $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$.

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Therefore $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$. ■

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Theorem (Local Sign-Preserving)

If $\lim_{x \rightarrow x_0} f(x) = A$ and $A > 0$ (resp. $A < 0$), then there exists a neighbourhood of x_0 where $f(x) > 0$ (resp. $f(x) < 0$).

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Hence $f(x) > A - \frac{A}{2} = \frac{A}{2} > 0$ in $\dot{U}(x_0, \delta)$. □

Corollary: Local Boundedness

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Proof.

Take $\varepsilon = 1$. $\exists \delta > 0$:

$$0 < |x - x_0| < \delta \implies |f(x) - A| < 1 \implies |f(x)| < |A| + 1.$$

Set $M = |A| + 1$. Then $|f(x)| \leq M$ for $x \in \dot{U}(x_0, \delta)$. □

(P32, Corollary)

Theorem 2: Order-Preserving Property

Theorem (Order-Preserving)

If in some deleted neighbourhood of x_0 :

$$f(x) \geq g(x) \quad \text{and} \quad \lim_{x \rightarrow x_0} f(x) = A, \quad \lim_{x \rightarrow x_0} g(x) = B,$$

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Remark on Strict Inequality

Caution

If the condition is changed to $f(x) > g(x)$ (strict), we can only conclude $A \geq B$, **not necessarily** $A > B$.

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Example (Counterexample)

$f(x) = x^2$, $g(x) = 0$ for all $x \neq 0$. Clearly $f(x) > g(x)$ for $x \neq 0$, but $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} g(x) = 0$.

Definition (Left Limit)

$$f(x_0^-) = \lim_{x \rightarrow x_0^-} f(x) = A \iff \forall \varepsilon > 0, \exists \delta > 0 : x \in (x_0 - \delta, x_0) \implies |f(x) - A| < \varepsilon.$$

Left Limit and Right Limit

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Definition (Right Limit)

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Theorem 3: Equivalence of Two-Sided and One-Sided Limits

Theorem (L/R Equivalence)

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(P30)

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(P30)

Practical Use

To check whether $\lim_{x \rightarrow x_0} f(x)$ exists:

- 1 Compute left limit $f(x_0^-)$.
- 2 Compute right limit $f(x_0^+)$.
- 3 If equal $\Rightarrow$ limit exists and equals that value.
- 4 If not equal $\Rightarrow$ limit does **not** exist.

Example 5: Does the Limit Exist?

Example

Let $f(x) = \begin{cases} x + 1, & x < 0 \\ x^2, & x \geq 0 \end{cases}$. Discuss whether $\lim_{x \rightarrow 0} f(x)$ exists.

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$$f(0^+) = \lim_{x \rightarrow 0^+} x^2 = 0.$$

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Solution.

Use Theorem 29 (L/R equivalence). **Right limit:**

$$f(0^+) = \lim_{x \rightarrow 0^+} x^2 = 0.$$

Left limit:

$$f(0^-) = \lim_{x \rightarrow 0^-} (x + 1) = 1.$$



Example 5 (continued)

Conclusion

Since $f(0^+) = 0 \neq 1 = f(0^-)$, the left and right limits differ.

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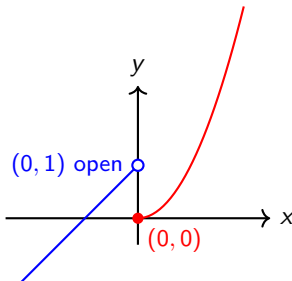
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Definition 2: Limit as $x \rightarrow \infty$

Definition ($x \rightarrow \infty$ Limit)

Let $f(x)$ be defined for $|x|$ greater than some positive number. If

$$\forall \varepsilon > 0, \exists X > 0 : |x| > X \implies |f(x) - A| < \varepsilon,$$

then A is the limit of $f(x)$ as $x \rightarrow \infty$:

$$\lim_{x \rightarrow \infty} f(x) = A \quad \text{or} \quad f(x) \rightarrow A \quad (x \rightarrow \infty).$$

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Equivalent Formulation

$$x < -X \quad \text{or} \quad x > X \quad \iff \quad A - \varepsilon < f(x) < A + \varepsilon.$$

Geometric Meaning: Horizontal Asymptote

II. independent variable tends to infinity when Limits of Functions

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Let function $f(x)$ positive number when
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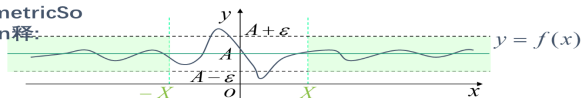
$$\lim_{x \rightarrow \infty} f(x) = A$$

或 $f(x) \rightarrow A$ (当 $x \rightarrow \infty$)

$$x < -X \text{ 或 } x > X$$

$$A - \varepsilon < f(x) < A + \varepsilon$$

**geometricSo
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line $y = A$ is curve $y = f(x)$'s horizontal asymptote

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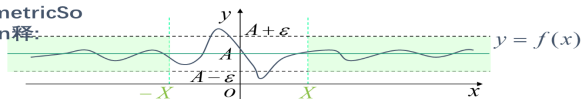
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Horizontal Asymptote

The line $y = A$ is a **horizontal asymptote** of the curve $y = f(x)$ if $\lim_{x \rightarrow \infty} f(x) = A$ (or $x \rightarrow \pm\infty$).

Special Cases: $x \rightarrow +\infty$ and $x \rightarrow -\infty$

Definition ($x \rightarrow +\infty$)

$$\lim_{x \rightarrow +\infty} f(x) = A \iff \forall \varepsilon > 0, \exists X > 0 : x > X \implies |f(x) - A| < \varepsilon.$$

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Definition ($x \rightarrow -\infty$)

$$\lim_{x \rightarrow -\infty} f(x) = A \iff \forall \varepsilon > 0, \exists X > 0 : x < -X \implies |f(x) - A| < \varepsilon.$$

Geometric Meaning

In both cases, $y = A$ remains a **horizontal asymptote** of $y = f(x)$.

Example 6: $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

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Prove $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$.

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For any $\varepsilon > 0$, we want $\frac{1}{|x|} < \varepsilon$, i.e. $|x| > \frac{1}{\varepsilon}$.

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Hence $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. ■

Example 6 (continued): Special Cases

Note

Similarly:

$$\lim_{x \rightarrow +\infty} \frac{1}{x} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

Examples of Horizontal Asymptotes

Example (Asymptote $y = 0$)

- $f(x) = \frac{1}{\sqrt{x}} \xrightarrow{x \rightarrow +\infty} 0$
- $g(x) = \frac{1}{\sqrt{1-x}} \xrightarrow{x \rightarrow -\infty} 0$

Both have horizontal asymptote $y = 0$.

Example (Asymptote $y = 1$)

- $f(x) = 1 - 2^{-x} \xrightarrow{x \rightarrow +\infty} 1$
- $g(x) = 1 + 2^x \xrightarrow{x \rightarrow -\infty} 1$

Both have horizontal asymptote $y = 1$.

Graphs of Horizontal Asymptotes

two kind special

case :

$$\lim_{x \rightarrow +\infty} f(x) = A \iff \forall \varepsilon > 0, \exists X > 0, \text{ 当 } x > X \text{ 时, 有 } |f(x) - A| < \varepsilon$$

$$\lim_{x \rightarrow -\infty} f(x) = A \iff \forall \varepsilon > 0, \exists X > 0, \text{ 当 } x < -X \text{ 时, 有 } |f(x) - A| < \varepsilon$$

geometric meaning: 直线 $y = A$ 仍是曲线 $y = f(x)$ 的渐近线.

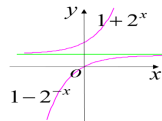
example:

$$\text{for } f(x) = \frac{1}{\sqrt{x}}, \quad g(x) = \frac{1}{\sqrt{1-x}}$$

all has horizontal asymptote $y = 0$;

$$\text{又 such as } f(x) = 1 - 2^{-x}, \quad g(x) = 1 + 2^{-x}$$

as, all has horizontal asymptote $y = 1$.



Summary: Limits of Functions

1 ε - δ / ε - X Definitions

- $x \rightarrow x_0$: $\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x - x_0| < \delta \Rightarrow |f(x) - A| < \varepsilon$
- $x \rightarrow \infty$: $\forall \varepsilon > 0, \exists X > 0 : |x| > X \Rightarrow |f(x) - A| < \varepsilon$

2 Properties:

- Uniqueness
- Local boundedness
- Local sign-preserving
- Order-preserving (monotonicity)
- L/R limit equivalence

3 Horizontal asymptote: $y = A$ if $\lim_{x \rightarrow \pm\infty} f(x) = A$

Question 1

If $\lim_{x \rightarrow x_0} f(x)$ exists, is $f(x)$ necessarily defined at $x = x_0$?

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Question 2

Let $f(x)$ and $g(x)$ both have limits as $x \rightarrow x_0$. Does $\lim_{x \rightarrow x_0} [f(x) \cdot g(x)]$ necessarily exist?